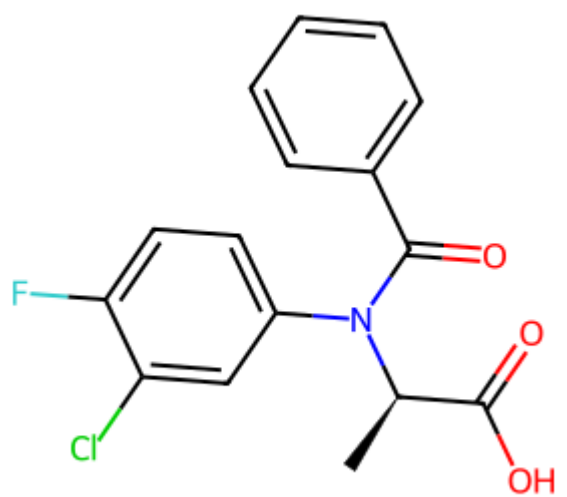
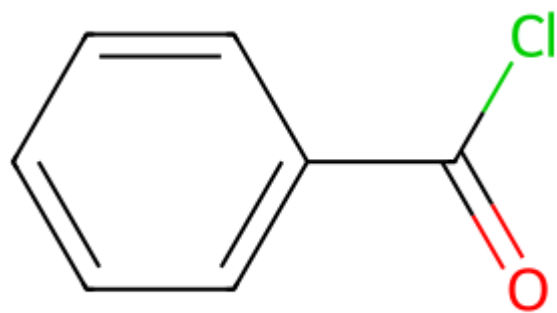




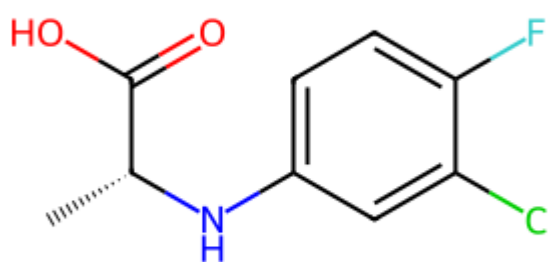
```
{'analysis_time': 0.0,  
'first_solution_time': 0.123986,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 6.009221268697241,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 6.009221268697241,  
'retro_star_value': 6.009221268697241}
```

Reaction

```
{'analysis_time': 0.123986,  
'first_solution_time': 0.888746,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 7.478353908126135,  
'retro_star_reaction_number': 7.478353908126135,  
'retro_star_rxn_cost': 1.3356012468043723,  
'retro_star_value': 7.478353908126135}
```



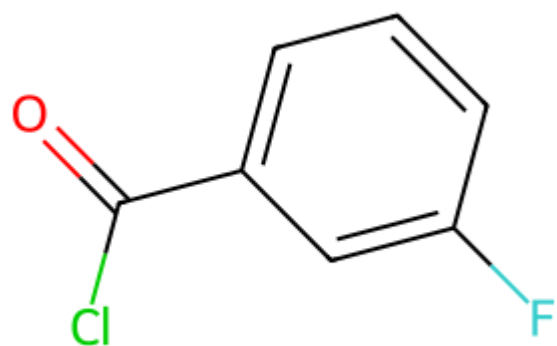
```
{'analysis_time': 0.123986,  
'first_solution_time': 0.888746,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 4.961845129926823,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 4.961845129926823,  
'retro_star_value': 7.478353908126135}
```



```
{'analysis_time': 0.123986,  
'first_solution_time': 0.424027,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 1.18090753139494,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 1.18090753139494,  
'retro_star_value': 7.478353908126135}
```

Reaction

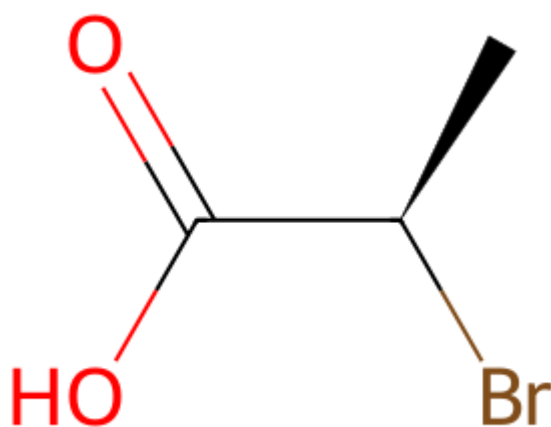
```
{'analysis_time': 0.888746,  
'first_solution_time': 0.888746,  
'num_calls_rxn_model': 10,  
'num_calls_value_function': 261,  
'retro_star_min_cost': 5.298317366548036,  
'retro_star_reaction_number': 5.298317366548036,  
'retro_star_rxn_cost': 5.298317366548036,  
'retro_star_value': 7.814826144747348}
```



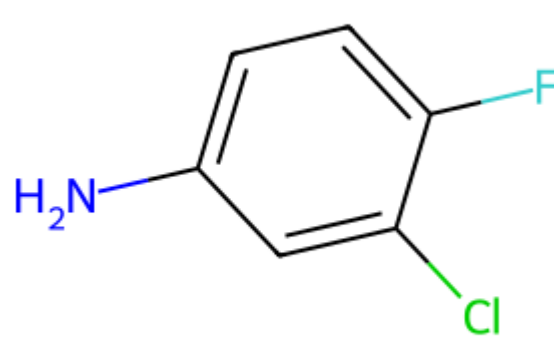
```
{'analysis_time': 0.888746,  
'first_solution_time': 0.888746,  
'num_calls_rxn_model': 10,  
'num_calls_value_function': 261,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.814826144747348}
```

Reaction

```
{'analysis_time': 0.424027,  
'first_solution_time': 0.424027,  
'num_calls_rxn_model': 4,  
'num_calls_value_function': 115,  
'retro_star_min_cost': 1.18090753139494,  
'retro_star_reaction_number': 1.18090753139494,  
'retro_star_rxn_cost': 1.18090753139494,  
'retro_star_value': 7.478353908126135}
```



```
{'analysis_time': 0.424027,  
'first_solution_time': 0.424027,  
'num_calls_rxn_model': 4,  
'num_calls_value_function': 115,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.478353908126135}
```



```
{'analysis_time': 0.424027,  
'first_solution_time': 0.424027,  
'num_calls_rxn_model': 4,  
'num_calls_value_function': 115,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.478353908126135}
```